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Dr. Jyh-Cherng Chern

Editor-in-Chief

International Journal of Pavement Research and Technology (IJPRT)

Subject: Submission of Original Research Article for Publication

Dear Dr. Chern,

I am pleased to submit our original research manuscript titled "**The Andean Thermal Pulse: Fourier Series Modeling of Rigid Pavement Behavior Under Extreme High-Andean Frost Cycles**" for consideration for publication as a research article in the *International Journal of Pavement Research and Technology*.

This study addresses a significant and overlooked gap in pavement engineering literature: the structural response of rigid pavements under high-altitude, diurnal frost cycles. While standard international literature focus heavily on seasonal (annual) freeze-thaw cycles in temperate climates, pavements in the high-Andean regions (above 3,200 m.a.s.l.) undergo complete curling and warping stress reversals every 24 hours due to extreme day-night temperature drops.

By isolating the dry winter season and utilizing a 5-harmonic discrete Fourier series analytical model calibrated with real hourly data from NASA POWER (MERRA-2) for Cusco, Peru, we proved that this high-frequency cycle triggers a progressive cumulative thermal fatigue mechanism. Furthermore, we implemented a mechanically derived Bradbury curling coefficient ($C = 0.2145$) based on the actual slab interaction with local subgrades, avoiding the arbitrary assumptions common in static pavement design catalogs. The model demonstrated high mathematical reliability with a Root Mean Square Error (RMSE) of 0.4044 °C.

We believe this work aligns perfectly with IJPRT's scope, particularly regarding your interest in regional pavement responses under extreme climatic variability and modern analytical validation.

This manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All dataset matrices and Python source codes have been made publicly available in a GitHub repository to ensure full scientific reproducibility. The authors declare no conflicts of interest.

Thank you very much for your time and for considering our manuscript.

Sincerely,

Anthoni Zevallos Pezo

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